Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/30/2007
PI Name:	Richter, Gregory Ph.D.		
Project Title:	NO Response to Gravistimulation in Arabidopsis Primary Root Tip		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	814-863-0278
Organization Name:	Penn State University Park		
PI Address 1:	Department of Biology		
PI Address 2:	208 Mueller Lab		
PI Web Page:			
City:	University Park	**State:**	PA
Zip Code:	16802	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Plants use the gravity vector to orient and optimize root growth patterns. Plants use the gravity vector to orient and optimize root growth patterns and root branching is a major determinant of this architecture. This investigation is focused on determining the factors controlling LR initiation in Arabidopsis primary root tips. Initial results indicate a strong correlation between LR initiation and the perception a change in the gravity vector. Thus, much of this project is geared toward determining the role of gravity in directing lateral root formation in response to tropic stimuli. Primary roots generating a curve of 45 degrees or greater due to tropic reorientation elicit lateral root induction on the convex side.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	The transport of auxin controls developmental events in plants. Here, we report that in addition to maintaining vacuolar pH, the H+-pyrophosphatase, AVP1, controls auxin transport and consequently auxin-dependent development. AVP1 overexpression results in increased cell division at the onset of organ formation, hyperplasia, and increased auxin transport. In contrast, avp1-1 null mutants have severely disrupted root and shoot development and reduced auxin transport. Changes in the expression of AVP1 affect the distribution and abundance of the P-adenosine triphosphatase and Pinformed 1 auxin efflux facilitator, two proteins implicated in auxin distribution. Thus, AVP1 facilitates the auxin fluxes that regulate organogenesis.
Bibliography Type:	Description: (Last Updated: 10/30/2007)
Articles in Peer-reviewed Journals	Li J, Yang H, Peer WA, Richter G, Blakeslee J, Bandyopadhyay A, Titapiwantakun B, Undurraga S, Khodakovskaya M, Richards EL, Krizek B, Murphy AS, Gilroy S, Gaxiola R. "Arabidopsis H+-PPase AVP1 regulates auxin-mediated organ development." Science. 2005 Oct 7;310(5745):121-5. PMID: 16210544 , Oct-2005